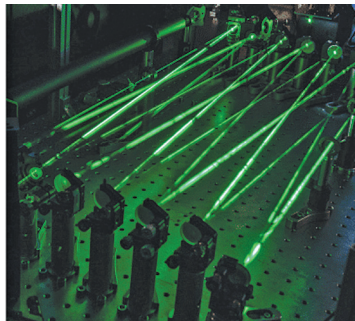




SWIR laser tracks emissions

Researchers at the Max Planck Institute for the Science of Light have developed a high-power ytterbium thin-disk laser integrated with an optical parametric oscillator to



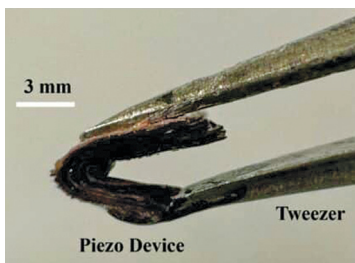
The optical parametric oscillator pump by ytterbium thin disk laser (Credit: MPL)

generate stable, high-intensity pulses in the short-wave infrared (SWIR) band. This system enables precise greenhouse gas tracking, mainly methane, which is 25 times more potent than CO₂. Traditional methods struggle due to spectral overlaps and water vapour interference, but

the SWIR band resolves these issues. The scalable laser detects real-time pollutant operating at megahertz rates with femtosecond pulses. Beyond environmental monitoring, it promises advancements in high-bandwidth earth-to-space optical communication systems.

A wearable generator to produce electricity

Researchers at the University of Waterloo have developed a wearable generator that uses vibrations and small body movements to generate electricity, potentially powering devices like smartphones and laptops sustainably. The technology employs the piezoelectric effect, where pressure on materials like crystals generates electrical energy. Traditional piezoelectric materials, often brittle and expensive, limit efficiency. To overcome this, the team combined



The new generator contains materials that are flexible, energy-efficient and relatively less expensive (Credit: University of Waterloo)

piezoceramics with flexible polymers to create piezoelectric nanogenerators (PENGs). These new PENGs are cost-effective, adaptable, and capable of generating high voltage, although increasing current output remains challenging.

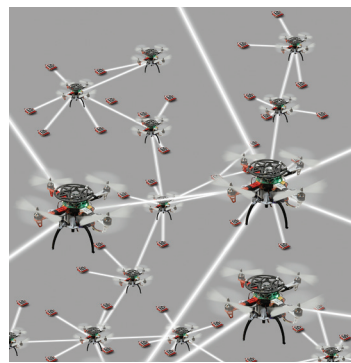
Optoelectronic device emulates human visual system

Researchers from Tsinghua University, China, have developed an optoelectronic device emulating the human visual system,

advancing in-sensor computing. The device integrates sensing, memory, and processing through a 1kb array of optoelectronic memristors (OEMs) configured with 128 × 8 cells and silicon CMOS circuits. These cells operate in three modes: electronic memristor, dynamic OEM, and non-volatile OEM (NV-OEM), enhancing energy efficiency and functionality. Constructed with Pd/TiO_x/ZnO/TiN layers, the system processes optical and electrical signals concurrently. It achieved 96.1% image recognition accuracy and reduced energy consumption by over 20 times compared to GPUs, enabling real-time visual processing in robotics, healthcare, and autonomous systems.

Advanced swarm robotics system

Researchers at Université Libre de Bruxelles have developed an advanced swarm robotics architecture, the



The self-organising nervous system allows a user to program an entire robot swarm as if it were a single robot with a reconfigurable morphology (Credit: Dr Mary Katherine Heinrich)

‘self-organising nervous system’ (SoNS), inspired by human neural networks. SoNS enables robots to dynamically self-organise into sub-swarms, optimising collaboration for tasks like sensing, movement, and mission planning. Acting as middleware, it blends centralised and self-organised control, allowing robots to adjust roles in ad-hoc hierarchies. Robots

communicate with nearby peers, minimising bottlenecks while balancing individual and collective behaviours. Tested with up to 250 robots, this system enhances scalability, flexibility, and fault tolerance.

PEI-TiO₂ layer boosts green hydrogen

Researchers from UNIST and the University of Zurich have developed a polyethyleneimine (PEI)-titanium dioxide (TiO₂) layer for metal oxide-based photoelectrodes, crucial in water-splitting electrochemical processes. The commercialisation of solar-driven green hydrogen production is advancing due to a protective material significantly enhancing photoelectrode durability. This layer selectively blocks electrons while enabling positively charged holes to drive oxidation, preventing corrosion and boosting performance. Applied to BiVO₄ photoanodes, it sustained water splitting for 400+ hours at 2.03mA/cm², a significant improvement over unprotected electrodes lasting only five hours.